

The Secret Family Tree

II PUC · Ch 4 · Principles of Inheritance and Variation

Starring Detective Ria 🕵️ & Dr. Gene 🧬

A BAIO storybook by Narendra Sir · Biology Lecturer



The Old Family Album

Detective Ria found an old album with photographs of four generations. Grandpa had dimples, Mother had none, but baby cousin Anu did! Great-grandma had attached earlobes, nobody in the next two generations did — and then Ria's brother did. "Why do traits skip generations?" she asked. Her friend Dr. Gene smiled. "The answer is **inheritance** — and it begins with a monk and his peas."

💡 **REMEMBER**

Inheritance is the process by which characters pass from parent to progeny; variation is the degree of difference between them.



Mendel’s Pea Garden

Between 1856 and 1863, **Gregor Mendel** crossed garden peas and counted thousands of plants. He chose seven pairs of **contrasting traits** such as tall and dwarf, and used **true-breeding** lines. "Mendel called the hidden units ‘factors’ — today we call them **genes**," said Dr. Gene. "Different forms of the same gene are **alleles**: T for tall, t for dwarf."

💡 **REMEMBER**

Mendel used statistical analysis and large samples, which made his conclusions reliable.



Dominant and Recessive

"When Mendel crossed tall TT with dwarf tt, all the F_1 plants were tall," Dr. Gene explained. "T is **dominant**; t is **recessive** — it hides in the presence of T." TT and tt are **homozygous**; Tt is **heterozygous**. Ria wrote: the letters are the **genotype**; what we see — tall or dwarf — is the **phenotype**.

💡 **REMEMBER**

A recessive trait appears only in the homozygous (tt) condition.



The Punnett Square

Dr. Gene drew a grid named after **Reginald Punnett**. For Tt × Tt, the gametes T and t met in four boxes: TT, Tt, Tt and tt. "Phenotypic ratio 3 tall : 1 dwarf, genotypic ratio **1:2:1**," Ria counted. "So if both parents are Aa, the chance of an aa child is one in four — 25 percent!" The **Punnett square** had cracked the first clue.

💡 REMEMBER

Law of Segregation: the two alleles of a gene separate during gamete formation, so each gamete gets only one.



PAGE 5 OF 8

The Test Cross

"But a tall plant could be TT or Tt. How do we know?" asked Ria. "Cross it with a dwarf tt plant — a **test cross**," said Dr. Gene. "All tall offspring means TT; half tall and half dwarf means Tt." In a dihybrid cross of seed colour and shape, Mendel got a 9:3:3:1 ratio, showing the **Law of Independent Assortment**.

💡 REMEMBER

Mendel's laws: Law of Dominance, Law of Segregation and Law of Independent Assortment.

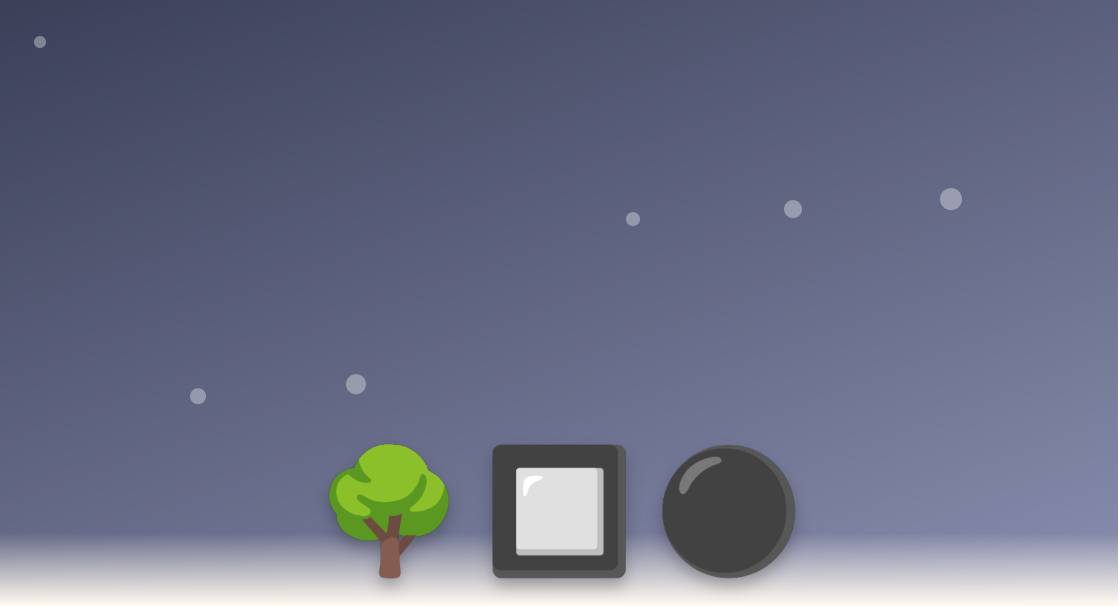


Beyond Mendel

In the snapdragon, red RR × white rr gave pink Rr flowers — **incomplete dominance**, with a 1:2:1 phenotypic ratio. Ria's blood group was AB: "Both I^A and I^B show together — **codominance**!" And because the ABO gene has three alleles, it is also an example of **multiple alleles**. One gene can even affect many traits — **pleiotropy**, as in phenylketonuria.

💡 **REMEMBER**

ABO blood groups: alleles I^A, I^B and i give six genotypes and four phenotypes (A, B, AB, O).



Reading the Pedigree

Ria drew the family as a **pedigree chart**: squares for males, circles for females, shaded for those with attached earlobes. "Look — two unaffected parents had an affected child. That is an **autosomal recessive** pattern," said Dr. Gene. "The same pattern is seen in **sickle-cell anaemia**. Traits on the X chromosome, like **haemophilia** and colour blindness, are **sex-linked**."

💡 **REMEMBER**

Sickle-cell anaemia: Glu is replaced by Val at the 6th position of the β-globin chain (GAG → GUG).



Mystery Solved

Ria closed the album. "Great-grandma was aa. Grandma and Grandpa were both **carriers** — Aa — so their children showed the dominant free earlobe. When two carriers had children again, one in four was aa. The trait never vanished; it was hiding!" Dr. Gene clapped. "That is why recessive traits skip generations. Case closed, Detective!"

💡 **REMEMBER**

Carriers are heterozygous individuals who do not show a recessive trait but can pass it on.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

inheritance p.1 · Gregor Mendel p.2 · contrasting traits p.2 · true-breeding p.2 · genes p.2 · alleles p.2 · dominant p.3 · recessive p.3 · homozygous p.3 · heterozygous p.3 · genotype p.3 · phenotype p.3 · Reginald Punnett p.4 · 1:2:1 p.4 · Punnett square p.4 · test cross p.5 · Law of Independent Assortment p.5 · incomplete dominance p.6 · codominance p.6 · multiple alleles p.6 · pleiotropy p.6 · pedigree chart p.7 · autosomal recessive p.7 · sickle-cell anaemia p.7 · haemophilia p.7 · sex-linked p.7 · carriers p.8



QUICK QUIZ

Do you remember?

1. What is the difference between genotype and phenotype?

- They are the same
- Genotype = genes; phenotype = visible trait
- Phenotype = genes; genotype = trait
- Genotype is only in plants

2. If both parents are Aa, what is the chance of an aa child?

- 0%
- 25%
- 50%
- 100%

3. What is an allele?

- A type of chromosome
- An alternative form of a gene
- A cell organelle
- A kind of mutation

4. ABO blood groups show:

- Incomplete dominance only
- Codominance and multiple alleles
- Complete dominance only
- Pleiotropy only

5. A test cross is a cross with a:

- Homozygous dominant parent
- Homozygous recessive parent
- Heterozygous parent
- Wild plant

Answers: 1-Genotype = genes; phenotype = visible trait · 2-25% · 3-An alternative form of a gene · 4-Codominance and multiple alleles · 5-Homozygous recessive parent